P3: Greedy Best-First Search

FIT1061 Pass Task — Week 3 — Searching Smarter, Not Harder

Greedy best-first search is the first step beyond blind search. The point of this task is to see how a simple rule for choosing what to expand next can change the behaviour of an algorithm in a meaningful way, even when the underlying search strategy stays the same.

Show us that you can

- You can explain why a greedy choice can be useful even when it is not guaranteed to be optimal.
- You can trace the behaviour of the frontier as the search proceeds.
- You can compare the results of greedy search with earlier search methods.

Give us

- A photographed hand trace of greedy best-first search — your single appended file.
- A completed notebook showing greedy best-first search, with your written reflection inside it.
- A short interpretation of what the algorithm does well and where it fails.

Resources

- Week 3 lecture slides and recording *(Moodle)*
- Hand-trace guide *(Moodle)*
- P3 starter notebook *(OnTrack)*
- P3_preread_priority_queues.pdf *(OnTrack + Moodle)*

Key sections in the task notebook

- Part 0: The hand-trace graph (P3.1)
- Part 1: Explore the priority queue
- Part 2: Implement distance functions (P3.2)
- Part 3: Implement greedy best-first search (P3.3)
- Part 4: Run on the small grid (5×5)
- Part 5: Run on the large grid (10×10)
- Part 6: Compare BFS vs Greedy Best-First
- Part 7: Understanding greedy best-first search (quiz)
- Part 8: Reflection — “Can a machine discover?” (P3.4)

Do the following

1. Review the lecture material and the pseudocode before coding.
2. Work through the notebook step by step and keep track of the frontier state.
3. Compare the result with BFS and DFS where relevant.
4. Upload your notebook plus a single photo/PDF of your hand trace; your reflection stays inside the notebook. Mark it ready for feedback.

! Acceptable use of Artificial Intelligence

AI tools are **discouraged** for implementation tasks at the Pass level. You may use AI for *explanation* (e.g., "explain what a queue is"), but not to generate code or write your reflection. Cite all AI use. The milestone discussion will probe for genuine understanding.
All submitted work must be your own.